Lecture: Non parameter Density Estimation:

(• Histogram / Parzen windows / KDE • KNN density estimator).

Histogram (直方图):

Sample $\{x_1 . x_N\}$

- consinder a region R
  - let $\pi = p(x \in R) = \int_R p(x)\,dx$
  - let k_R = # samples in R
  - MLE = $\hat{\pi} = \frac{k_R}{N}$

Assume R is small ∴ $P(x)$ 在R内几乎不变

- $\hat{\pi} \approx p(x) V_R$. V_R 是区域体积,

Solve: $\hat{p}(x) \approx \frac{\hat{\pi}}{V_R} = \frac{k_R}{N V_R}$

- How to choose "R"?

1) keep V_R fixed, let k_R vary => (kernal density estimate (KDE), Parze windows)

2) keep k_R fixed let V_R vary => (KNN density estimator (really bad).

① KDE (kernal density estimate), Parzen windows:

- Let R be the d-dimensional hypercube. slide length h. => $V_R = h^d$.

- introduce a window (unit hypercube):

$$k(x) = \begin{cases} 1 & |x_i| \le \frac{1}{2} \text{ where } \forall i \in \{1..d\}. \\ 0, & \text{otherwise}. \end{cases}$$

⇓ $k\left(\frac{x - x_i}{h}\right) = \begin{cases} 1, & \text{if } x \text{ falls inside a hypercube of slide length } h \text{ centered at } x_i \\ 0, & \text{otherwise} \end{cases}$

⇓ # points near to x:

$$k = \sum_{i=1}^{N} k\left(\frac{x - x_i}{h}\right)$$

- Hence: $\hat{p}(x) = \frac{k_R}{N V_R} = \frac{1}{N h^d} \sum_{i=1}^{N} k\left(\frac{x - x_i}{h}\right)$

- kernel functions:

constrains: $k(x) \ge 0$. $\int k(x) = 1$

- Box kernel: $k(x) = \begin{cases} 1 & |x_i| \le \frac{1}{2} \\ 0 \end{cases}$

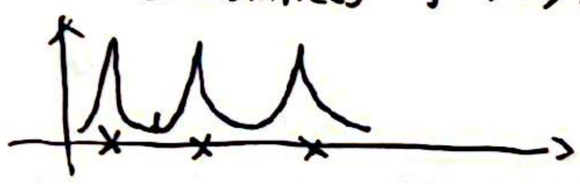
- sphere kernel: $k(x) = \begin{cases} 1/\text{Volumn}(\text{d-space unit ball}) & \|x\|^2 < 1 \\ 0, & \text{otherwise} \end{cases}$

- Gaussian kernel: $k(x) = \frac{1}{(2\pi)^{d/2}} e^{-\frac{1}{2}\|x\|^2}$

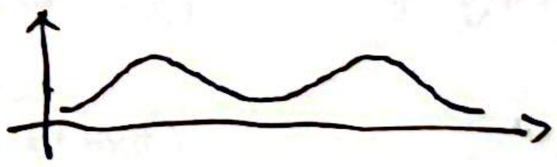
$$\Rightarrow \hat{p}(x) = \frac{1}{N h^d} \sum_{i=1}^{N} \frac{1}{(2\pi)^{d/2}} e^{-\frac{1}{2}\left\|\frac{x - x_i}{h}\right\|^2} = \frac{1}{N} \sum_{i=1}^{N} N(x \mid x_i, h^2 I)$$

- Bandwidth Parameter h.
  - h controls smoothness of $\hat{p}(x)$.

- h small (noisy)



- h large (blurry)



Convergence Analysis:

- $\hat{p}(x)$ will converge to true $p(x)$?
  - $\hat{p}(x) \to p(x)$: ① $\lim_{N\to\infty} E[\hat{p}(x)] = p(x)$ ② $\lim_{N\to\infty} Var(\hat{p}(x)) = 0$

- (scaled kernel): $\tilde{k}(x) = \frac{1}{h^d} k(\frac{x}{h})$ → scales width.
  ↑ scales

$$\Rightarrow \hat{p}(x) = \frac{1}{N}\sum_{i=1}^{N} \tilde{k}(x-x_i)$$

- Mean: $E[\hat{p}(x)] = E_{x_i}\left[\frac{1}{N}\sum_{i=1}^{N} k(x-x_i)\right]$

$$= \int p(z)\tilde{k}(x-z)\,dz = p(x) * \tilde{k}(x)$$

To get $E[\hat{p}(x)] \Rightarrow p(x)$, we need $\tilde{k}_0(x) = \delta(x) = \lim_{h\to 0} \tilde{k}(x)$

$\tilde{k}_h(x) \to \delta(x)$ 当 $h \to 0$. (dirac delta).

- Variance: $Var(\hat{p}(x)) = var\left(\frac{1}{N}\sum_i \tilde{k}(x-x_i)\right)$

$Var(\hat{p}(x)) \le \frac{1}{Nh^d}[\max_x k(x)][E(\hat{p}(x))]$ to get small variance

need N large. h large.

* h controls the tradeoff betwen bia & var.

$h \to 0 \Rightarrow$	bias $\to 0$, var > 0
$h \to \infty \Rightarrow$	bias > 0 var $\to 0$

How to select h:
- According to the physical process of setting of x_i
- set h to maximize LL of held of set.

Mean shift Algorithm:

任务与聚类方式: • Find mode of $\hat{p}(x)$

idea: 1) start with initial point $\hat{x}$

2) use gradient ascent ($\hat{x} \leftarrow \hat{x} + \lambda \nabla \hat{p}(x)$)

3) $\hat{x}$ converges to a mode.

- repeat different $\hat{x}$ to get all modes
- clustering: 收敛到一个mode的点到一类.

- Consider only radially symmetric kernels.

$k(x) = \alpha \bar{k}(\|x\|^2)$ → kernel profile.

Ex: Gaussian:

$k(x) = \frac{1}{(2\pi)^{d/2}} e^{-\frac{1}{2}\|x\|^2} \Rightarrow \alpha = \frac{1}{(2\pi)^{d/2}}$

$\bar{k}(r) = e^{-\frac{1}{2}r}$

Density estimate: $\hat{p}(x) = \frac{\alpha}{Nh^d}\sum_{i=1}^{N} \bar{k}\left(\left\|\frac{x-x_i}{h}\right\|^2\right)$

gradient: $\bar{g}(r) = -\bar{k}'(r)$

$$\nabla\hat{p}(x) \propto \frac{2\alpha}{Nh^d}\left(\sum_{i=1}^{N}\bar{g}\left(\left\|\frac{x-x_i}{h}\right\|^2\right)\right)\underbrace{\left(\frac{\sum_{i=1}^{N} x_i \bar{g}\left(\left\|\frac{x-x_i}{h}\right\|^2\right)}{\sum_{i=1}^{N}\bar{g}\left(\left\|\frac{x-x_i}{h}\right\|^2\right)} - x\right)}_{m(x)}$$

mean shife vector

$m(x) - x$

M2

Gradient Ascent : $\hat{x}^{(k+1)} = \hat{x}^{(k)} + \lambda \nabla \hat{p}(\hat{x}^{(k)})$

use a adapted step size : $\lambda = \frac{1}{\hat{g}(x)}$

$$\Rightarrow \hat{x}^{(k+1)} = \hat{x}^{(k)} + \frac{1}{\hat{g}(x)} \hat{g}(x)\, m(\hat{x}^{(k)})$$

- $$\hat{x}^{(k+1)} = \frac{\sum_{i=1}^{N} x_i\, \bar{g}\left(\left\|\frac{x^k - x_i}{h}\right\|^2\right)}{\sum_{i=1}^{N} \bar{g}\left(\left\|\frac{x^k - x_i}{h}\right\|^2\right)} \rightarrow x^{(k+1)} = m(x^{(k)})$$

if the kernel profile (monotonically decreasing and convex) ,) converge to a stationary point.